



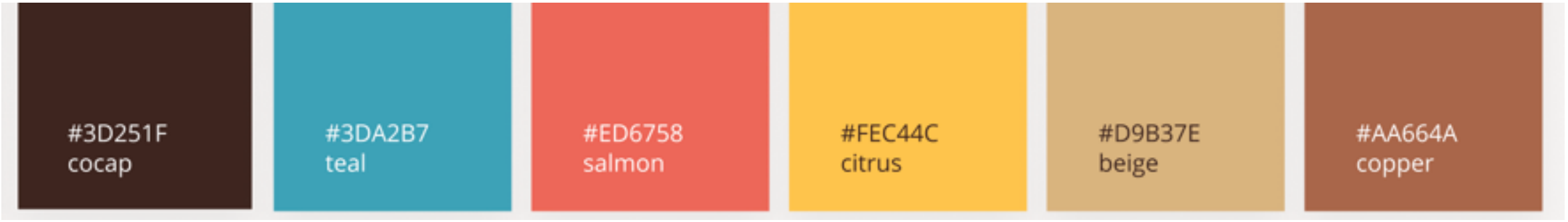
compute vector-
Jacobian products on-the-fly

works with any parameterization

leveraging

existing, memory-efficient ops like log-sum-exp

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#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Autodiff: advantages

- **Avoids Explicit Jacobian Computation:** Autodiff frameworks usually **compute vector-Jacobian products on-the-fly**, avoiding full Jacobian construction.
- **GPU-Accelerated and ML—Friendly:** Autodiff integrates seamlessly with modern libraries (e.g., PyTorch, JAX), enabling efficient batched computation and backpropagation on GPU.
- **Handles arbitrary models:** **works with any parameterization** of $a(\theta)$, $x(\theta)$, or $c_\phi(x, y)$, including deep networks, regardless of complexity.
- **Exploits optimized ops:** Autodiff traces gradients through full Sinkhorn loop, **leveraging existing, memory-efficient ops like log-sum-exp** already built into DL libraries.
- **Flexible Accuracy-Speed Tradeoff:** Tune number of Sinkhorn iterations or use `.detach()`-based approximations to balance gradient fidelity and computational efficiency.

Stop Grad Trick